

Shreyan Jain

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OVERVIEW

Firmware and Embedded Software Engineer with 3+ years of experience in bootloader, BIOS, BSP, and low-level firmware development for ARM-based platforms. Proficient in C, Python, and embedded system design, with strong exposure to pre-silicon bring-up, platform initialisation, device interfaces (I2C, SPI, UART), and system integration. Adept at designing build pipelines, automating firmware flows, and enabling cross-platform firmware deployment.

TECHNICAL SKILLS

Languages: C, C++17, Python, Bash (proficient); Rust, Go (familiar)

Embedded Linux & BSP: Yocto (kirkstone, dunfell), Linux kernel (LKM, device drivers), U-Boot, eMMC boot & partitioning, A/B image updates, OTA firmware update

Firmware & Protocols: ARM Cortex-M, BLE 5.x, RS-485, I2C, SPI, TCP/IP, MQTT; CAN 2.0B, FreeRTOS (familiar)

Vision & Acceleration: GStreamer, NVIDIA Argus, MIPI CSI, V4L2, CUDA, TensorRT

Platforms: TI AM625 / AM62-SIP, AMD Zynq UltraScale+ (ZU3EG), NVIDIA Jetson Orin Nano

Debugging & Testing: GDB, JTAG, Valgrind, ASan, UBSan; libFuzzer, AFL, VectorCAST (familiar)

Tools: Git, CMake, Make, Docker, Jenkins, Jira, Perforce, Qt

EXPERIENCE

Systems Software Engineer

Jan 2025 – Present

Astrogate Labs

Bangalore, India

- Developed firmware update infrastructure using half-duplex RS-485 for an 8-node test jig. This involved creating a custom framing system with CRC-16 for every 256B chunk, along with an A/B image scheme that automatically rolls back in case of a boot failure or power loss, tested for frame corruption and power cuts.
- developed and managed a Yocto (kirkstone, Dunfell) cross-compilation process that takes x86_64 and targets AArch64 for Zynq UltraScale+ MPSoC (ZU3EG) platforms.
- Developed a real-time computer vision pipeline using the NVIDIA Jetson Orin Nano Super
- Processed MIPI CSI camera streams using GStreamer, Argus, and shared CPU/GPU buffers, minimizing memory copies and reducing frame latency, for tracking and beacon acquisition

Software Engineer

Jul 2023 – Dec 2024

Raptee HV

Chennai, India

- Designed and owned full BLE 5.x App layer implementation on top of the vendor-provided BLE controller SDK (C, ARM Cortex-M).
- Developed OTA firmware update functionality and cloud connectivity for a custom AM625-based ECU platform.
- Developed and optimised OTA scripts for automated firmware deployment and system recovery
- Implemented eMMC boot, partition configuration, partition switching, and firmware upgrade mechanisms.
- Developed Yocto BSPs for AM62-SIP boards.

Software Engineer Intern

Jan 2023 – Jul 2023

Senpronics

Bangalore, India

- Developed Data Acquisition software and algorithms for various sensors, using Qt's C++ framework and TCP/IP protocols.
- Implemented logging, and numerical computations on the acquired data using boost libraries for downstream teams' analysis/alerting tools and tracking signal-to-noise ratio.

PROJECTS AND ACHIEVEMENTS

- Compiler:** A basic yet scalable compiler written in Python that compiles a dialect of BASIC to C. Supports numerical variables, basic arithmetic, if statements, while loops, print text and numbers, input numbers, labels and goto, comments. [GitHub](#)
- DNS Client:** Resolves the DNS for a particular domain through CLI commands, along with additional info, completely written using Rust, can take in and parse arguments accordingly. [GitHub](#)
- Git:** Created a version control tool that can create a repo, commit, and push itself to GitHub, written in Python and implemented using its libraries. [GitHub](#)
- Snake Game:** Created a snake game using only HTML, CSS and JavaScript, which even has a scoring system and can run in any browser. [GitHub](#)
- Qualified for ICPC West Asia Regionals 2020, got selected from the top 700 colleges (in India) to do so. [Certificate](#)
- Google Kickstart Round G – got 1704th rank worldwide.

EDUCATION

Bachelor of Engineering in Computer Science

October 2022

St Josephs Engineering College

Mangalore, India